“CUSTOMER SEGMENTATION ANALYSIS USING PYTHON”

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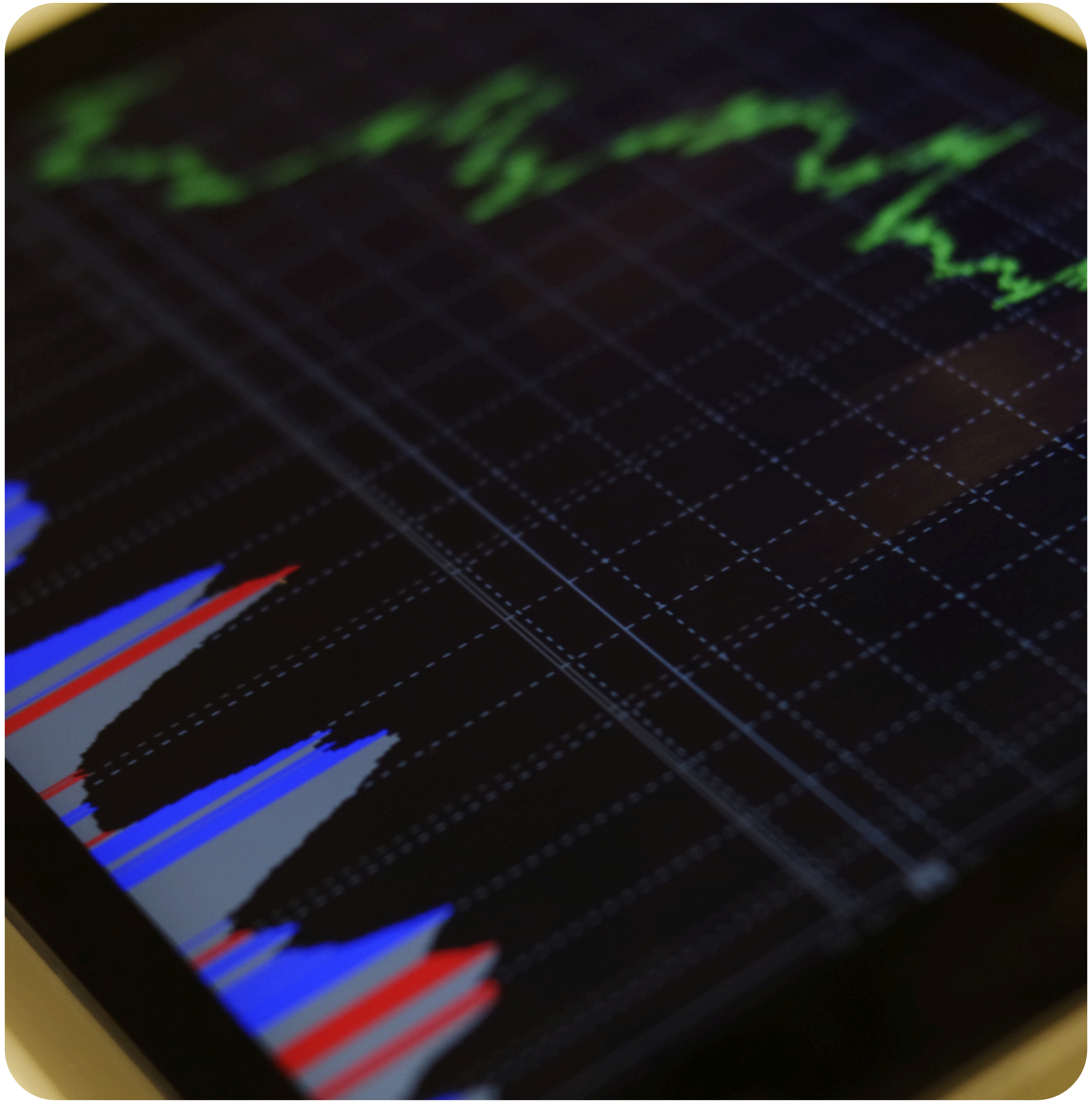


OBJECTIVE

“To identify high-value customer segments and analyze spending behavior based on age and income”

Tech Stack: Python (Pandas, Matplotlib)

[1]:
```python
import pandas as pd
import matplotlib.pyplot as plt

df = pd.read_csv("Mall_Customers.csv")
df.head()
```

[1]:

	CustomerID	Gender	Age	Annual Income (k$)	Spending Score (1-100)
0	1	Male	19	15	39
1	2	Male	21	15	81
2	3	Female	20	16	6
3	4	Female	23	16	77
4	5	Female	31	17	40

[2]:
```python
df.info()
```

```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 200 entries, 0 to 199
Data columns (total 5 columns):
 #   Column                  Non-Null Count  Dtype
---  ------                  --------------  -----
 0   CustomerID              200 non-null    int64
 1   Gender                  200 non-null    object
 2   Age                     200 non-null    int64
 3   Annual Income (k$)      200 non-null    int64
 4   Spending Score (1-100)  200 non-null    int64
dtypes: int64(4), object(1)
memory usage: 7.9+ KB
```

"The dataset is perfectly clean with 200 complete entries and no missing values, making it ready for immediate segmentation analysis using the core income and spending metrics"

[3]:
```python
df.isnull().sum()
```

[3]:
```
CustomerID                0
Gender                    0
Age                       0
Annual Income (k$)        0
Spending Score (1-100)    0
dtype: int64
```

[4]:
```python
df.describe()
```

[4]:

	CustomerID	Age	Annual Income (k$)	Spending Score (1-100)
count	200.000000	200.000000	200.000000	200.000000
mean	100.500000	38.850000	60.560000	50.200000
std	57.879185	13.969007	26.264721	25.823522
min	1.000000	18.000000	15.000000	1.000000
25%	50.750000	28.750000	41.500000	34.750000
50%	100.500000	36.000000	61.500000	50.000000
75%	150.250000	49.000000	78.000000	73.000000
max	200.000000	70.000000	137.000000	99.000000

“The customers' average age is 39 with a mean annual income of $60.56k and a balanced spending score of 50.2, indicating a middle-aged, moderate-spending target audience”

[5]:
```python
print("Total Customers:", len(df))

print("Average Spending:", df['Spending Score (1-100)'].mean())
```

```
Total Customers: 200
Average Spending: 50.2
```

[6]:
```python
high = df[df['Spending Score (1-100)'] > 70]
print("High spenders:", len(high))
```

```
High spenders: 54
```

[7]:
```python
df['Age Group'] = pd.cut(df['Age'],
                        bins=[18,25,35,50,70],
                        labels=['18-25','26-35','36-50','50+'])

df['Age Group'].value_counts()
```

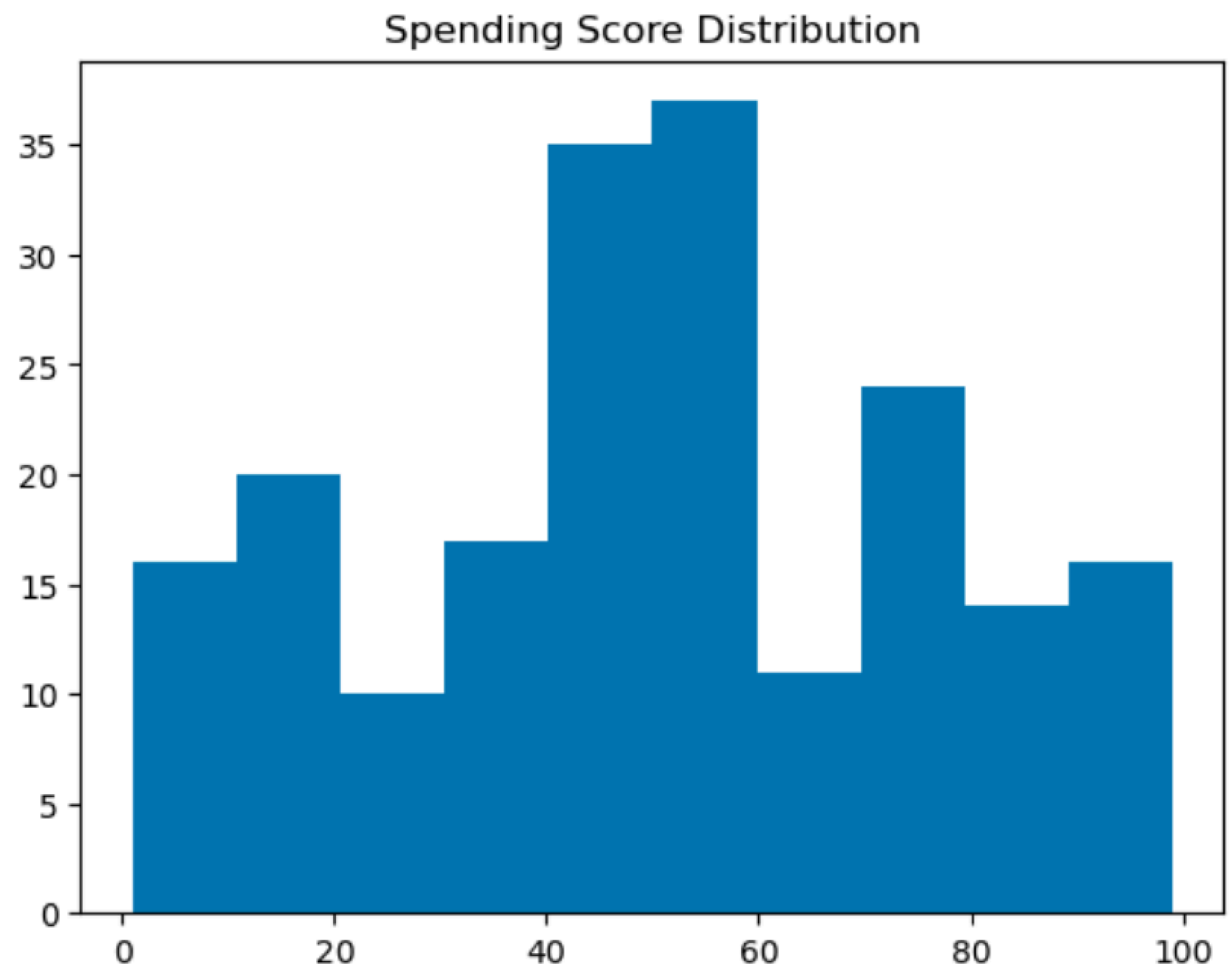
[7]:
```
Age Group
36-50    62
26-35    60
50+      40
18-25    34
Name: count, dtype: int64
```

"Out of 200 total customers, 27% are high spenders, with the largest segment being the 26-50 age group, which accounts for over 60% of the mall's footfall"

[8]:

```python
import matplotlib.pyplot as plt

plt.hist(df['Spending Score (1-100)'])
plt.title("Spending Score Distribution")
plt.show()
```



"The spending score shows a major peak between 40 and 60, indicating that the bulk of customers have moderate spending habits rather than extreme high or low patterns"

```
[9]: plt.scatter(df['Annual Income (k$)'], df['Spending Score (1-100)'])
plt.title("Income vs Spending")
plt.show()
```

“The scatter plot clearly identifies five distinct customer clusters, showing that high income doesn't always mean high spending, which allows the mall to create specific strategies for each group”

THANK YOU

"This analysis provides a roadmap for data-driven marketing and improved customer engagement."